Design and Implementation of a Zero Trust–Based Micro-VPN for Campus Networks

Abhirami C U
Department of Computer Science and Engineering
Rajagiri School of Engineering and Technology
Kochi, Kerala, India

Alan Suresh Joseph
Department of Computer Science and Engineering
Rajagiri School of Engineering and Technology
Kochi, Kerala, India

Aparna Velickakathu Sunil
Department of Computer Science and Engineering
Rajagiri School of Engineering and Technology
Kochi, Kerala, India

Aryasree Nambiar
Department of Computer Science and Engineering
Rajagiri School of Engineering and Technology
Kochi, Kerala, India

Mr. Sandy Joseph
Department of Computer Science and Engineering
Rajagiri School of Engineering and Technology
Kochi, Kerala, India

***Abstract*—Traditionally, campus networks have employed perimeter security solutions like firewalls and Virtual Private Networks (VPNs). Although these solutions ensure encrypted communication and the first level of authentication, they tend to trust users after gaining access. This trust model poses risks, especially in an academic setting where various parties share common resources. This paper describes the design and implementation of a lightweight Micro-VPN solution for campus networks that adopts the Zero Trust model. The proposed design combines multi-factor authentication, role-based access control, adaptive trust score calculation, and micro-level segmentation. A simulated campus network setup for a student management portal proves the viability of the proposed model. Experimental analysis confirms the prevention of unauthorized lateral movement, continuous policy enforcement, centralized auditing, and negligible performance overhead. The results show that the Zero Trust paradigm can be efficiently applied to academic institutions without requiring enterprise-grade infrastructure.**

***Index Terms*—Zero Trust Architecture, Micro-VPN, Campus Security, Role-Based Access Control, Trust Score, Network Segmentation**

I. Introduction

Most learning institutions have campus networks that employ perimeter security models. In these models, users are required to log in via VPN gateways or firewalls, after which they are allowed access to internal networks. While data is encrypted, there is implicit trust that is granted after the user has logged in. This approach can be dangerous, especially when thousands of users log in daily to access centralized systems.

Learning management system portals store highly confidential information such as academic performance, attendance, financial transactions, and personal identifiers. If the user's credentials are compromised, attackers can use the implicit trust feature to gain access to internal services or laterally move across systems.

Zero Trust Architecture (ZTA) is a model that eliminates implicit trust in favor of continuous verification [1]. In this model, instead of trusting users based on their initial login credentials, all requests for access are subject to identity, role, and context-based policies. Nevertheless, most enterprise ZTA implementations are expensive and complex.

This paper proposes a lightweight Micro-VPN solution based on Zero Trust architecture tailored for campus networks.

II. Related Work

NIST Special Publication 800-207 defines Zero Trust Architecture and stresses the need to remove implicit trust within networks [1]. Microsegmentation and identity-driven access control have been extensively researched as methods to minimize attack surfaces [5], [7].

Recent work has proposed adaptive trust scoring models that dynamically assess user behavior [4], [6]. Enterprise-level implementations show that Zero Trust Architecture can greatly improve security posture [11]. Nevertheless, systematic reviews have pointed out cost and complexity hurdles for smaller institutions like universities [12].

These works inform the need to build a simplified and more scalable Zero Trust model for academic settings. The application of Zero Trust principles at scale came after the Google BeyondCorp model [7], which abandoned VPN perimeter security. Enterprise-level work has shown that Zero Trust Architecture can greatly improve attack surfaces and breach containment [11]. AI-based extensions have further improved anomaly scoring and behavioral risk analysis [9], [10]. Nevertheless, systematic reviews have shown that cost and

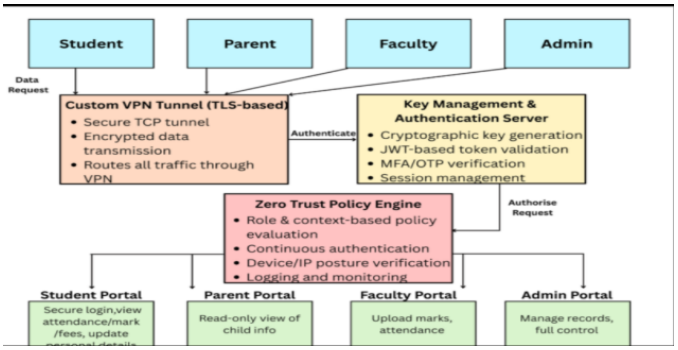

Fig. 1. Proposed Zero Trust VPN Architecture

complexity hurdles remain substantial for smaller institutions [12].

III. Proposed Architecture

The proposed system combines the principles of Zero Trust with secure VPN communication, cryptographic security, adaptive access control, and continuous trust evaluation. Unlike the traditional security paradigm, in which users inside the perimeter are trusted implicitly after successful authentication, Zero Trust requires continuous authentication of all access requests [1].

The design is intended to ensure that every request is dynamically validated, regardless of its source. The design is in line with the latest concepts of Zero Trust models and microsegmentation strategies, as described in recent literature [2], [5]. The entire process is broken down into five main stages, as described below.

A. Stage 1: User Request Initiation

The process starts with a user trying to gain access to a secured academic resource. The user logs into the system using a secure login page and provides primary authentication details in terms of username and password.

However, as opposed to traditional systems, gaining access is not immediate after successful verification of the authentication details. As per the Zero Trust model, no user, computer, or network location can ever be trusted [1]. Each and every request is considered to be malicious until it is verified. The login request is securely logged and then sent to the next security level for encrypted transmission.

This is more secure than traditional systems, which can face issues of unauthorized access due to compromised authentication details, and is in line with current enterprise Zero Trust implementation models [3], [4].

B. Stage 2: Secure VPN Tunnel Cryptographic Layer

After the initiation of the user request, a secure VPN tunnel is created between the client and the organizational server. This Micro-VPN architecture is distinct from the conventional VPN in that it only provides access to particular services instead of granting access to the entire internal network [5].

The VPN layer is responsible for ensuring that all communications are encrypted using robust cryptographic methods. Symmetric encryption is employed for secure data transfer because of its efficiency, and asymmetric encryption is employed for secure key exchange. These practices adhere to the principles of secure networking [8].

The cryptographic layer protects sensitive academic data from interception, tampering, and replay attacks. Even if the network infrastructure is compromised, secure communication through encryption guarantees the confidentiality and integrity of the data. This layer is the secure transport base of the proposed system and adheres to the guidelines for secure communication in the Zero Trust model [1].

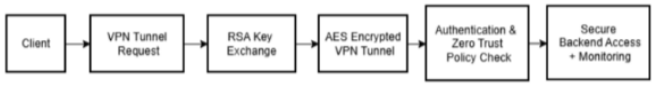

Fig. 2. Secure VPN Tunnel and Cryptographic Layer

C. Stage 3: Key Management and Authentication

In this phase, the secure key generation and management processes are established. The cryptographic session keys are dynamically generated and securely exchanged using the authenticated key exchange mechanisms to avoid any unauthorized key disclosure.

To add an extra layer of identity verification, the Multi-Factor Authentication (MFA) process is incorporated into the system. After the password authentication is successfully performed, the user is required to undergo an OTP-based verification process. The OTP is dynamically generated and securely delivered via a trusted communication channel. Only after the successful OTP authentication is the authentication process deemed complete.

The incorporation of the MFA process greatly minimizes the threats posed by credential-based attacks and phishing attacks. The concept of Zero Trust research is highly reinforced by the need for MFA as a mandatory security mechanism [2], [6]. Moreover, the adaptive authentication methods are increasingly being suggested in the emerging Zero Trust models [9], [10].

This multi-layered authentication process ensures that the use of compromised credentials alone does not lead to any unauthorized access to the system.

D. Stage 4: Zero Trust Evaluation and Request Validation

After authentication, the request undergoes strict Zero Trust policy evaluation. Instead of granting blanket access, the system evaluates multiple contextual parameters, including:

- User role
- Device information
- Access location
- Time of request
- Historical behavior patterns

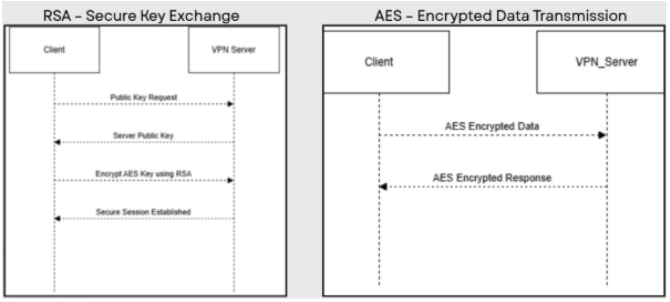

Fig. 3. Key Management and Authentication

These contextual cues are also used in deciding whether the request is in line with the organizational access policies. The system implements role-based access control (RBAC) to ensure that the principle of least privilege is followed [3], [5].

Additionally, microsegmentation at the service level ensures that users are only able to access resources that are specifically allowed for their role, thereby inhibiting any unauthorized lateral movement within the network [7]. This is in line with contemporary approaches to Zero Trust, which decouple authentication from authorization decisions [1].

The assessment and testing phase is dynamic and policy-driven to ensure that authentication status is not the sole determinant of continued access.

E. Stage 5: Trust Score Evaluation, Access Enforcement and Security Monitoring

In the final stage, the system calculates a dynamic trust score based on user behavior, contextual risk factors, and request patterns. Trust-based adaptive models are increasingly recognized as effective mechanisms in Zero Trust ecosystems [9], [11].

$$T = 100 - (\alpha * F_{login}) - (\beta * F_{otp}) - (\gamma * V_{rbac}) - (\delta * R_{limit})$$

where,

F_{login}: The number of failed login attempts,

F_{otp}: The number of failed OTP verification attempts,

V_{rbac} : Value of role-based access control violations, and

R_{limit} : The number of rate-limit threshold violations.

The weights used in the implementation are as follows:
$\alpha = 25$ for failed login attempts,
$\beta = 20$ for failed OTPs,
$\gamma = 15$ for violations of RBAC
$\delta = 10$ for rate limit violations.

After calculation, the trust score is normalized so that it stays within proper values:

$$T = \max(0, \min(T, 100))$$

Fig. 4. Calculation of Trust Value

If the calculated trust value exceeds the set threshold, access is granted with permissions strictly corresponding to the user's assigned role. If the trust value shows moderate risk, restricted access or re-authentication can be required. If the trust value goes below the acceptable level, the request will be denied and marked for administrative analysis.

Every user action is recorded in a centralized monitoring system. Real-time monitoring and auditing improve transparency and response to incidents [12]. Continuous assessment eliminates internal threats, privilege escalation, and lateral movement attacks, thus reducing the attack surface area substantially compared to traditional perimeter-based VPN solutions [11].

By these five phases, the proposed Micro-VPN architecture provides secure communication, strict policy enforcement, adaptive trust validation, and continuous monitoring—thus forming a complete Zero Trust access management system that is applicable in an academic setting.

IV. Experimental Results and Performance Evaluation

This section presents the experimental setup, evaluation metrics, and performance analysis of the proposed Zero Trust based secure VPN framework.

A. Experimental Setup

The system was designed and developed using a secure client-server architecture. The client interface was designed to support the simulation of different user roles such as an administrator, employee, and guest user. The server side handled the user authentication process, OTP verification, trust score computation, and logging.

The experiments were conducted in a controlled network environment to evaluate the accuracy of authentication, reliability of access control, adaptability of trust scores, and system overhead.

B. Evaluation Metrics

To measure the efficiency of the proposed Micro-VPN-based Zero Trust architecture, different parameters have been considered. These parameters are in line with the Zero Trust model, as described by NIST [1], and also in line with the principles of the recent studies performed on Zero Trust [2], [3].

The parameters are listed in Table I.

Authentication Accuracy assesses the ability of the system to distinguish between genuine users and malicious actors. Comprehensive identity verification is the cornerstone of all Zero Trust architectures [1], [4].

OTP Verification Reliability quantifies the reliability of the multi-factor authentication tier. Multi-factor authentication techniques are strongly advised in Zero Trust architectures to resist brute-force and replay attacks [2], [6].

Role-Based Access Control (RBAC) Enforcement ensures that users are limited to the services they are entitled to, based on their roles. The concept of least privilege and micro-segmentation is given importance in the architecture of Zero Trust and BeyondCorp [5], [7].

Trust Score Responsiveness focuses on the efficiency of the system in recalculating trust scores based on the changes in

TABLE I
EVALUATION METRICS FOR MICRO-VPN ARCHITECTURE

Metric	Definition	Evaluation Objective
Authentication Accuracy	Tests the system's ability to authenticate valid users and reject unauthorized users	To test the efficiency of identity verification systems in the Micro-VPN gateway
OTP Verification Reliability	Tests the success rate of correct one-time password entry in normal and abnormal conditions	To ensure resistance to brute-force and replay attacks
Role-Based Access Control Enforcement	Tests users' ability to access services only as per their respective roles	To verify the consistent enforcement of least-privilege policies
Trust Score Responsiveness	Tests the speed and accuracy of trust score updates	To evaluate dynamic adaptation to behavioral changes
Unauthorized Access Detection Rate	Percentage of abnormal access attempts successfully prevented	To measure effectiveness of Zero Trust enforcement
System Response Time	Time taken to authenticate and authorize/deny access	To ensure acceptable latency
User Session Continuity	Tests the ability to maintain secure sessions under continuous verification	To ensure usability under strict policies
Overall User Experience	Qualitative assessment of responsiveness and usability	To confirm practical feasibility

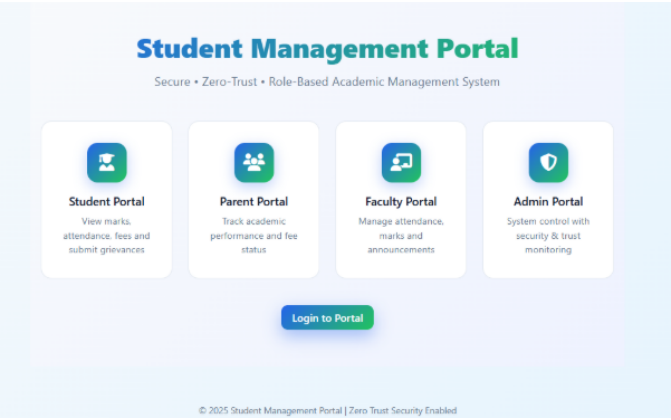

Fig. 5. Portal Dashboard

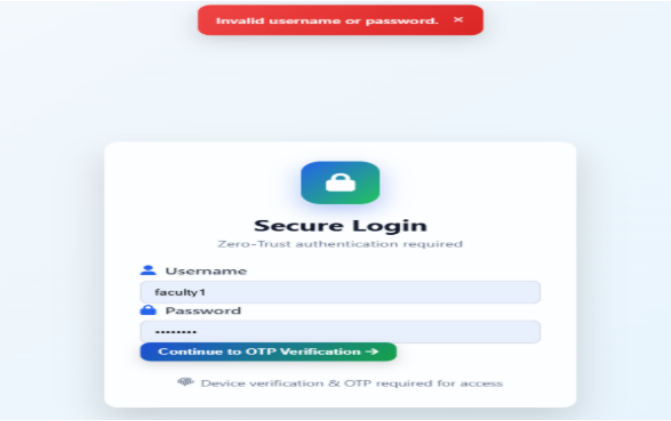

Fig. 6. OTP Entry Interface

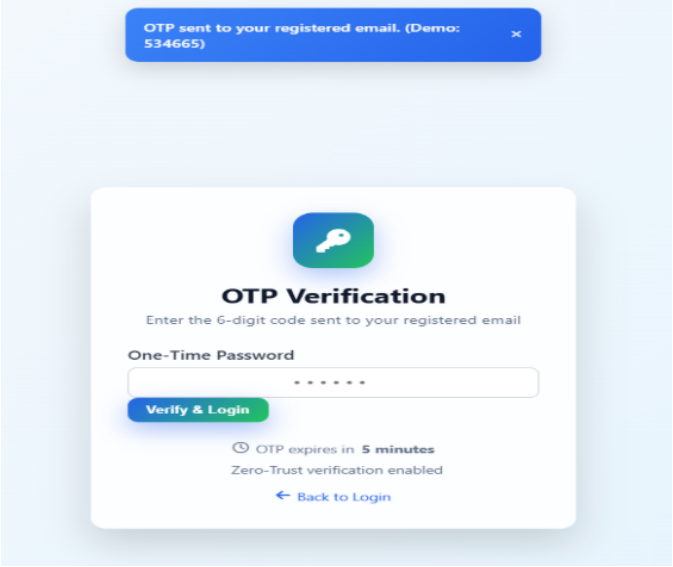

Fig. 7. OTP Verification

user behavior. Recent works show the importance of adaptive and AI-optimized trust models in contemporary Zero Trust architectures [9], [10].

Unauthorized Access Detection Rate calculates the rate at which the system successfully identifies and prevents malicious access attempts. Experimental studies show the effectiveness of Zero Trust enforcement in reducing the attack surface area and improving enterprise security [11], [12].

System Response Time assesses the impact of the security layers on the system's performance. Network security solutions must ensure that protection does not come at the cost of performance [8].

User Session Continuity calculates the system's capability to ensure continuous trust validation for user sessions. This is a fundamental requirement in the architecture of Zero Trust models [1], [3].

Overall User Experience offers qualitative feedback on the system's usability and responsiveness, which is essential for its implementation in academic settings [2], [11].

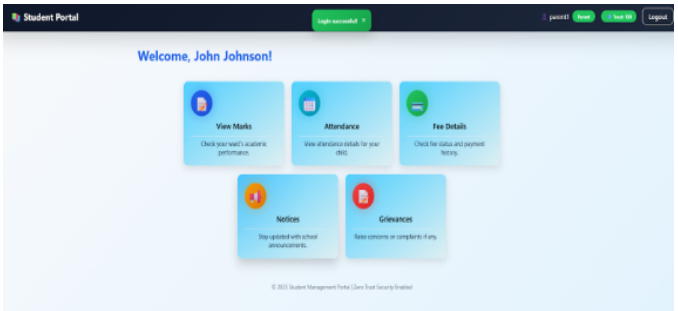

Fig. 8. Student Portal Dashboard After Successful Authentication

Together, these metrics validate the security effectiveness, adaptability, enforcement consistency, and operational efficiency of the proposed Zero Trust Micro-VPN architecture.

C. Authentication and OTP Verification Results

The authentication module was very reliable with minimal false rejection rates. The addition of OTP verification provided an additional level of security without any delay. The combination of passwords and OTPs ensured that there were no unauthorized login attempts.

D. Role-Based Access Control Enforcement

The role-based access control module was implemented correctly in the system. Administrative users had all access privileges, while employee and guest users had restricted access to predefined resources. There was no privilege escalation during the testing phase.

E. Trust Score Evaluation and Adaptive Access Control

The trust score module was dynamic in nature and changed according to user behavior, providing access privileges accordingly. Login attempts from unknown IP addresses were considered to be malicious and resulted in low trust scores and restricted access. Users with normal behavior had high trust scores and continuous access.

F. Prevention of Unauthorized Lateral Movement

The Zero Trust model was highly successful in preventing lateral movement. Even after successful authentication, the user was not free to roam across the network. The user was limited to certain segments of the network according to his/her role and trust score. Any attempt to access unauthorized resources was denied and logged.

G. Logging, Monitoring, and Auditability

All attempts to access resources, whether successful or unsuccessful, were logged in a systematic manner. Real-time monitoring allowed the system to detect any unusual activity, such as multiple unsuccessful login attempts.

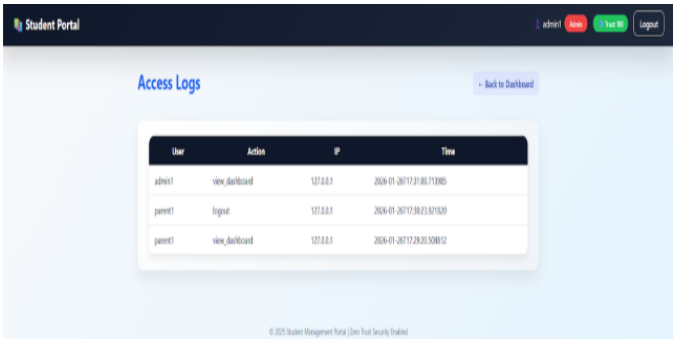

Fig. 9. Admin portal displaying centralized security logs

H. Performance Overhead Analysis

Although multiple security layers were implemented, the overall system overhead remained within acceptable limits. The average response time was below two seconds, indicating that enhanced security did not significantly degrade usability.

The cryptographic layer and trust evaluation modules introduced minor computational overhead, but this trade-off is justified by the improved security posture.

I. Discussion of Results

The experimental results show that the integration of Zero Trust concepts with secure VPN communication has a significant positive impact on the security of the organization. The system successfully prevented unauthorized access, reduced lateral movement, and maintained high authentication accuracy.

The adaptive trust score model performed well in adjusting access levels. The results show that effective security solutions can be achieved without a significant impact on performance.

The proposed framework is a scalable and efficient solution for secure remote access.

V. Conclusion and Future Work

This project has offered a Micro-VPN solution based on the Zero Trust model, specifically designed for campus networks. Through the combination of multi-factor authentication, RBAC enforcement, adaptive trust evaluation, and micro-level service segmentation, this solution removes implicit trust and prevents lateral movement.

The experimental results have shown the effectiveness of this solution in access control, continuous validation, centralized auditing, and performance overhead. This solution is scalable and can be applied in an institutional setting.

Future improvements will focus on the integration of AI-powered behavioral anomaly detection and dynamic policy adjustment.

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